

Internship report

The Riemann-Hilbert correspondence

by
Damien J_UNGER

supervised by
Annette H_UBER-K_LAWITTER

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I Prerequisites

I.1 Category theory

Definition I.1.

A category $\mathcal{C}$ is given by the data of :

1. a class $\text{Ob}(\mathcal{C})$, called the *class of objects* of $\mathcal{C}$,
2. for every couple of objects (x, y) of $\mathcal{C}$, a class $\text{Hom}_{\mathcal{C}}(x, y)$, called the *class of morphisms* from x to y ,
3. for every triplet of objects (x, y, z) of $\mathcal{C}$, a function

$$\begin{aligned} \text{comp}_{(x,y,z)} : \text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(y, z) &\longrightarrow \text{Hom}_{\mathcal{C}}(x, z) \\ (f, g) &\longmapsto g \circ f \end{aligned}$$

satisfying the following properties :

- a. (*Associative law*) for every quartet of objects (w, x, y, z) of $\mathcal{C}$, if $(f, g, h) \in \text{Hom}_{\mathcal{C}}(w, x) \times \text{Hom}_{\mathcal{C}}(x, y) \times \text{Hom}_{\mathcal{C}}(y, z)$, then $(h \circ g) \circ f = h \circ (g \circ f)$,
- b. (*Identities*) for every object x of $\mathcal{C}$, there exists an element 1_x of $\text{Hom}_{\mathcal{C}}(x, x)$, so that for every couple of objects (w, y) and every morphisms $(f, g) \in \text{Hom}_{\mathcal{C}}(w, x) \times \text{Hom}_{\mathcal{C}}(x, y)$, then $1_x \circ f = f$ and $g \circ 1_x = g$.

$g \circ f$ is called the *composition* of g with f .

Remark I.1.

1. We will use the following notations : $x \in \text{Ob}(\mathcal{C})$ if x is an object of $\mathcal{C}$, $f : x \rightarrow y$ if f is an element of $\text{Hom}_{\mathcal{C}}(x, y)$.
2. If $f : x \rightarrow y$, x is called the *domain* or *source* of f and y the *codomain* or *target* of f .
3. For our purpose, it is sufficient to consider that for every $(x, y) \in \text{Ob}(\mathcal{C})^2$, $\text{Hom}_{\mathcal{C}}(x, y)$ is a set. In this case, the category is said to be *locally small*.

Definition I.2.

Let $\mathcal{C}$ be a category.

A category $\mathcal{D}$ is said to be a *subcategory* of $\mathcal{C}$ if the following are satisfied :

1. every object of $\mathcal{D}$ is an object of $\mathcal{C}$,
2. for every $(x, y) \in \text{Ob}(\mathcal{D})^2$, then $\text{Hom}_{\mathcal{D}}(x, y) \subseteq \text{Hom}_{\mathcal{C}}(x, y)$,
3. for every $x \in \text{Ob}(\mathcal{D})$, then $1_x \in \text{Hom}_{\mathcal{D}}(x, x)$,
4. for every $(x, y, z) \in \text{Ob}(\mathcal{D})^3$, every $(f, g) \in \text{Hom}_{\mathcal{D}}(x, y) \times \text{Hom}_{\mathcal{D}}(y, z)$, then $g \circ f \in \text{Hom}_{\mathcal{D}}(x, z)$.

$\mathcal{D}$ is said to be a *full subcategory* of $\mathcal{C}$ if in addition, for every $(x, y) \in \text{Ob}(\mathcal{D})^2$, then $\text{Hom}_{\mathcal{D}}(x, y) = \text{Hom}_{\mathcal{C}}(x, y)$.

I.2 Sheaf theory

In the following, X is a topological space and $\mathcal{T}_X$ will denote its topology category.

Definition I.3.

Let $\mathcal{C}$ be a category (often $\mathcal{C} = \mathcal{S}, \mathcal{A}, \mathcal{R} \dots$).

A *presheaf* of $\mathcal{C}$ is a contravariant functor $\mathcal{F} : \mathcal{T}_X \rightarrow \mathcal{C}$.

Explicitly, it is the data of :

1. for every open set U of X , an object $\mathcal{F}(U)$ of $\mathcal{C}$,
2. for every couple of open sets $V \subseteq U$ of X , a morphism $\text{res}_{U,V} : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ verifying :
 - a. for every open set U of X , then $\text{res}_{U,U} = \text{id}_U$,
 - b. for every open sets $U \subseteq V \subseteq W$ of X , then $\text{res}_{U,V} \circ \text{res}_{V,W} = \text{res}_{U,W}$.

The objects $\mathcal{F}(U)$ are often called *sections of $\mathcal{F}$ over U* , and for avoiding confusion, will we denoted $\Gamma(U, \mathcal{F})$.

Remark I.2.

If the context is clear, we will denote $\text{res}_{U,V}(f) = f|_U$ (the notation is justified in the next lines).

Example I.3.

- 1) The sheaf of real valued continuous functions $C^0(-, \mathbb{R})$ with classical restriction functions $\text{res}(U, V)(f) = f|_U$.
- 2) If X has an additional structure, the above example can be further specified : if X is a differentiable manifold, we can consider $C^\infty(-, \mathbb{R})$, if X is a complex manifold, $\mathcal{O}(-, \mathbb{C}) \dots$
- 3) If C is an object of $\mathcal{C}$, the constant presheaf $\underline{C}$. We will be intersted in $\underline{\mathbb{Z}}$ or $\underline{\mathbb{C}}$.
- 4) The sheaf of real valued bounded functions $B(-, \mathbb{R})$.

Presheaves comes from a category point of view, so we need morphisms :

Definition I.4.

A *morphism of presheaves* (over the same space X) $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a natural transformation.

More explicitly, it is the data, for each $U \in \mathcal{T}_X$, of a morphism $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ of $\mathcal{C}$ such that the following diagram commutes for every $U \subseteq V$:

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \\ \text{res}_{U,V}^{\mathcal{F}} \downarrow & & \downarrow \text{res}_{U,V}^{\mathcal{G}} \\ \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \end{array}$$

Definition I.5.

A *sheaf* $\mathcal{F}$ is a presheaf satisfying the extra conditions for any open cover $\{U_i, i \in I\}$ of an open set U of X :

3. for every $f, g \in \mathcal{F}(U)$ such that $\forall i \in I, f|_{U_i} = g|_{U_i}$, then $f = g$.
4. for every $(f_i) \in \prod_{i \in I} \mathcal{F}(U_i)$ such that $\forall (i, j) \in I^2, f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$, then there exists $f \in \mathcal{F}(U)$ with $f|_{U_i} = f_i$ for any $i \in I$.

Remark I.4.

For $U = \emptyset$, it says that $\mathcal{F}(\emptyset)$ is the terminal object of $\mathcal{C}$ (if such one exists). Indeed, we can take the open covering of $\emptyset$ by a family indexed by $\emptyset$ itself, and we then have an arrow $\mathcal{F}(\emptyset) \rightarrow$

Example I.5.

- 1) and 2) are indeed sheaves.
- 3) is not, because of the previous remark.
- 4) is not either, with the counter example $\mathbb{R} = \bigcup_{n \in \mathbb{N}} (-n, n)$ and $f : x \mapsto x$.
- If X is Hausdorff, x_* is a point of X , (S, s) a pointed set, the *skyscraper sheaf* is given by
$$\text{sky}_{x_*}(S) : U \mapsto \begin{cases} S & \text{if } x_* \in U \\ s & \text{otherwise.} \end{cases}$$

Definition I.6.

We will denote by PSh_X , respectively Sh_X , the category of presheaves, respectively the full subcategory of sheaves, over X .

One can ask what looks like the behavior of the sheaf very locally, on a point. Since a singleton is in general not a point, we need the following definition.

Definition I.7.

Let $\mathcal{F}$ be a (pre)sheaf.

We define the *stalk* at $x \in X$ as :

$$\mathcal{F}_x := \{(U, s) \mid x \in U, U \in \mathcal{T}_X, s \in \mathcal{F}(U)\} / \sim,$$

where $(U, s) \sim (V, t) \Leftrightarrow (\exists W \subset U \cap V) : s|_W = t|_W$.

Remark I.6.

From a categorical point of view, it is also the direct limit : $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$, where the limit is taken over open set U .

It turns out that this localization principle can be used to transform a presheaf into a sheaf in a natural way (in the sense of what follows).

Definition I.8.

Let $\mathcal{F}$ be a presheaf.

We define for every open set U

$$\mathcal{F}^\sharp(U) = \left\{ s : U \rightarrow \prod_{x \in U} \mathcal{F}_x \mid \left\{ \begin{array}{l} \forall x \in U, s(x) \in \mathcal{F}_x \\ \exists V \subset U, \exists t \in \mathcal{F}(V) \text{ s.th. } \forall y \in V, s(y) = t(y) \end{array} \right\} \right\}.$$

$\mathcal{F}^\sharp$ is called the *sheafification* of $\mathcal{F}$.

Proposition I.1.

$\mathcal{F}^\sharp$ is a sheaf.

Example I.7.

$\mathbb{R}^\sharp$ is the sheaf of locally constant functions, more precisely the sheaf of continuous functions $C^0(-, \mathbb{R})$, where $\mathbb{R}$ is equipped with the discrete topology.

Proposition I.2.

$((-)^{\sharp}, \text{Forget})$ is an adjoint pair of functors between PSh_X and Sh_X .

PROOF

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□

Example I.8.

Suppose that $\mathcal{C}$ admits kernels and cokernels (this assumption holds whenever kernels and cokernels are written).

For $\varphi : \mathcal{F} \rightarrow \mathcal{G}$, we define $\text{Ker}(\varphi)(U) := \text{Ker}(\varphi_U)$, $\text{Im}(\varphi)(U) := \text{Im}(\varphi_U)$ and $\text{Coker}(\varphi)(U) := \text{Coker}(\varphi_U)$.

The first one is always a sheaf, but the two others are not in general. In the following, $\text{Im}(\varphi)$ or $\text{Coker}(\varphi)$ will always denote the sheafification of these presheaves.

Proposition I.3.

A complex $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ is exact if, and only if the induced complex on stalks $0 \rightarrow \mathcal{F}_x \rightarrow \mathcal{G}_x \rightarrow \mathcal{H}_x \rightarrow 0$ is exact for every $x \in X$.

PROOF

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□

Remark I.9.

In general, it is not equivalent of the exactness of the induced sequence over every open set U . One could take as a counterexample the exponential sequence on $\mathbb{C}$:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathbb{Z} & \rightarrow & \mathcal{O}_{\mathbb{C}} & \rightarrow & \mathcal{O}_{\mathbb{C}}^* \rightarrow 0, \\ & & & & f & \mapsto & \exp(2\pi i f) \end{array}$$

that is not exact over $\mathbb{C}^*$, because the map $\mathcal{O}_{\mathbb{C}^*} \rightarrow \mathcal{O}_{\mathbb{C}^*}^*$ is not surjective : the map $z \mapsto z$ does not belong to the image since $\log$ is not well defined over $\mathbb{C}^*$.

Proposition I.4.

Let $X = \cup_{i \in I} U_i$ an open covering of X . For every $i \in I$, let $\mathcal{F}_i$ a sheaf on U_i . Assume that for every $i \neq j \in I$, there exists an isomorphism

$$\theta_{i,j} : \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j},$$

satisfying the condition $\theta_{j,k} \circ \theta_{i,j} = \theta_{i,k}$ on $U_i \cap U_j \cap U_k$ for every triplet of distinct elements of I .

Then, there exists a sheaf $\mathcal{F}$ on X such that for every $i \in I$, $\mathcal{F}|_{U_i} = \mathcal{F}_i$. Moreover, this sheaf is unique, up to unique isomorphism.

PROOF

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□

I.3 Complex geometry

I.3.1 First definitions

Definition I.9.

Let X be a differentiable manifold of dimension $2n$.

An *holomorphic atlas* is a set $\{(U_i, \varphi_i), i \in I\}$ where $X = \cup_{i \in I} U_i$ is an open cover and $\varphi : U_i \rightarrow \mathbb{C}^n$ are homeomorphism on their image, such that for all $(i, j) \in I^2$, the *transition map* $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \rightarrow \varphi_j(U_i \cap U_j)$ is holomorphic.

Such a couple (U_i, φ_i) is called a *local coordinate system*, or just *coordinates*.

We say that two holomorphic atlases $\{(U_i, \varphi_i), i \in I\}$ and $\{(\tilde{U}_j, \tilde{\varphi}_j), j \in J\}$ are *equivalent* if for all $(i, j) \in I \times J$, $\tilde{\varphi}_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap \tilde{U}_j) \rightarrow \tilde{\varphi}_j(U_i \cap \tilde{U}_j)$.

Definition I.10.

A *complex manifold of dimension n* is a real manifold X of dimension $2n$ endowed with an equivalence class of holomorphic atlases.

Remark I.10.

In the following, unless explicitly stated otherwise, the complex manifold will be endowed with an representant of the equivalent class, denoted by $\{(U_i, \varphi_i), i \in I\}$.

Definition I.11.

Let X be a complex manifold and $f : X \rightarrow \mathbb{C}$.

We say that f is *holomorphic* if for all local coordinates (U, φ) , $f \circ \varphi : \varphi(U) \rightarrow \mathbb{C}$ is holomorphic.

Definition I.12.

Let X be a complex manifold.

We denote by $\mathcal{O}_X$ the sheaf of holomorphic function on X , ie $\mathcal{O}_X : U \mapsto \{f : U \rightarrow \mathbb{C}, f \text{ holomorphic}\}$.

Remark I.11.

$\mathcal{O}_{X,x}$ will denote the stalk of $\mathcal{O}_X$ at x . It coincides with the germs of holomorphic functions at x . Via local coordinates (U, φ) , with $\varphi(x_*) = 0$ for some $x_* \in U$, we have an isomorphism $\mathcal{O}_{X,x} \simeq \mathcal{O}_{\mathbb{C}^n,0}$. It is an integral domain, and we denote by $Q(\mathcal{O}_{X,x})$ the quotient field associated.

More generally, if the codomain is also a complex manifold, we have the following definition :

Definition I.13.

Let X and Y be complex manifolds and $f : X \rightarrow Y$ continuous.

We say that f is *holomorphic* if for all local coordinates (U, φ) on X and (U', φ') on Y , $\varphi' \circ f \circ \varphi : \varphi(U \cap f^{-1}(U')) \rightarrow \varphi'(U')$ is holomorphic.

Definition I.14.

Let X be a complex manifold.

A *meromorphic function* on X is a map $f : X \rightarrow \bigsqcup_{x \in X} Q(\mathcal{O}_{X,x})$ so that, for all $x_* \in X$, there exists an open neighborhood $U \subset X$ of x_* and $g, h \in \mathcal{O}_U$ verifying : $(\forall x \in U), f_x = \frac{g}{h}$.

The sheaf of meromorphic functions is denoted $\mathcal{K}_X$.

Remark I.12.

If $U \subset X$ is an connected open subset, then $\mathcal{K}_X(U)$ is a field, and if X itself is connected, $\mathcal{K}(X) = \mathcal{K}_X(X)$ is called the function field of X .

I.3.2 Vector bundle**Definition I.15.**

Let X be a complex manifold and r a positive integer.

A *holomorphic vector bundle* or rank r on X is a complex manifold E equipped with a holomorphic map $\pi : E \rightarrow X$ such that :

1. for all $p \in X$, the *fiber* $E_p := \pi^{-1}(p)$ is a complex vector space of dimension r ,
2. there exists an open covering $X = \bigcup_{i \in I} U_i$ and biholomorphic maps $\psi_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{C}^r$, called the *trivialization maps*, satisfying :
 - a. the following diagram commutes :

$$\begin{array}{ccc}
 \pi^{-1}(U_i) & \xrightarrow{\psi_i} & U_i \times \mathbb{C}^r \\
 \searrow \pi & & \swarrow \text{pr}_1 \\
 & U_i &
 \end{array}$$

b. The induced map $E_p \rightarrow \mathbb{C}^r$ for any $p \in U_i$ is $\mathbb{C}$ -linear.

If $r = 1$, we say that (E, π) is a *line bundle*.

Remark I.13.

The trivialization maps induce linear isomorphisms $\psi_{i,j} = \psi_i \circ \psi_j^{-1} : U_i \cap U_j \rightarrow \text{GL}_r(\mathbb{C})$, and the collection $\{\psi_{i,j}, i, j \in I\}$ is called a cocycle.

In particular, it verifies the conditions :

- i. $\psi_{i,i} = \text{id}_{U_i}$,
- ii. $\psi_{j,k} \circ \psi_{i,j} = \psi_{i,k}$.

Proposition I.5.

Let X be a complex manifold and let r be a positive integer. The categories $\text{VectBun}^r(X)$ and $\text{LocFree}_r(X)$ are equivalent.

PROOF

We define

$$\begin{aligned} \mathcal{F} : \text{VectBun}_r(X) &\longrightarrow \text{LocFree}_r(X) \\ E &\longmapsto \mathcal{F}_E \end{aligned},$$

where $\mathcal{F}_E$ is the sheaf of sections of E . It is well defined by ??.

It is a functor : given a morphism of vector bundles $f : E_1 \rightarrow E_2$, we define for each open set U of X the function

$$\begin{aligned} \varphi_U : \mathcal{F}_1(U) &\longrightarrow \mathcal{F}_2(U) \\ s &\longmapsto f \circ s \end{aligned}.$$

φ clearly defines a morphism of sheaves.

In order to construct a quasi-inverse, let $\mathcal{F} \in \mathcal{S}_X$ and $\psi : \mathcal{F}(U_i) \rightarrow \mathcal{O}_{U_i}^{\oplus r}$ local trivialisations on an open cover $\{U_i, i \in I\}$ of X .

□

I.3.3 Connections

Definition I.16.

Let $\mathcal{E}$ be a locally free $\mathcal{O}_X$ -modules.

A *connection on $\mathcal{E}$* is a $\mathbb{C}$ -linear application $\nabla : \mathcal{E} \rightarrow \mathcal{E} \otimes_{\mathcal{O}_X} \Omega_X^1$ satisfying the *Liebniz relation* : for any local sections s and local function f , then

$$\nabla(fs) = df \otimes s + f\nabla(s).$$

In particular, it induces, for each integer $k > 0$, a $\mathbb{C}$ -linear map

$$\begin{aligned} \nabla^{(k)} : \mathcal{E} \otimes_{\mathcal{O}_X} \Omega_X^k &\longrightarrow \mathcal{E} \otimes_{\mathcal{O}_X} \Omega_X^{k+1} \\ s \otimes \omega &\longmapsto (-1)^k \nabla s \wedge \omega + s \otimes d\omega \end{aligned}$$

A connection ∇ is said to be *flat* (or *integrable*) if the *curvature* $K := \nabla^{(1)} \circ \nabla$ is zero.

Remark I.14.

In dimension 1, we have $\Omega_X^2 = 0$, so all connections are flat.

Example I.15.

- 1) d is a connection on $\mathcal{O}_X$ by identifying $\mathcal{O}_X \otimes_{\mathcal{O}_X} \Omega_X^1 \simeq \Omega_X^1$. It induces a connection on $\mathcal{O}_X^r$ acting coordinate-wise.
- 2) For every $\omega \in \Omega_X^1$, the map

$$\begin{aligned} d + \omega : \mathcal{O}_X &\longrightarrow \Omega_X^1 \\ f &\longmapsto df + \omega f \end{aligned}$$

is a connection. More generally, if ω is a $r \times r$ matrix with coefficients in Ω_X^1 , the map $d + \omega$ is a connection on $\mathcal{O}_X^r$.

The following result says that these are the only ones on $\mathcal{O}_X^r$, and then the only ones locally on $\mathcal{E}$.

Lemma I.6.

Every connection ∇ on $\mathcal{O}_X^r$ is written $d + \omega$ for some ω as above.

PROOF

For every $f \in \mathcal{O}_X$ and $h \in \mathcal{O}_X^r$, we have by the Liebniz rule :

$$(\nabla - d)(fh) = f\nabla(h) + df \otimes h - f dh - df \otimes h = f(\nabla - d)h.$$

Thus, $\nabla - d : \mathcal{O}_X^r \rightarrow \Omega_X^1$ is $\mathcal{O}_X$ -linear and it's the multiplication by a matrix ω .

□

Remark I.16.

Connections are a reformulation of (systems) of differential equations, locally on a vector bundle.

Let $\nabla = d + \omega$ a connection on $\mathcal{O}_X^r$ (it is of this form by the previous lemma). We choose local coordinates $z^1, \dots, z^n$ on X . Then we can write

$$\omega(z) = \sum_{j=1}^n A_j(z) dz^j.$$

For a f holomorphic on the chart, we then have :

$$\nabla f = \sum_{j=1}^n \left(\frac{\partial f}{\partial x^j} + A_j f \right) dz^j,$$

and $\nabla f = 0$ is equivalent to :

$$\begin{cases} \frac{\partial f}{\partial z^1}(z) = A_1(z)f(z), \\ \vdots \\ \frac{\partial f}{\partial z^n}(z) = A_n(z)f(z), \end{cases}$$

which is a system of *partial* differential equations.

Definition I.17.

We define the category $\text{LocFree}^\nabla(X)$ with

- locally free $\mathcal{O}_X$ -modules equipped with flat connections as objects,
- morphisms of $\mathcal{O}_X$ -modules $\varphi : \mathcal{E}_1 \rightarrow \mathcal{E}_2$ making

$$\begin{array}{ccc} \mathcal{E}_1 & \xrightarrow{\nabla_1} & \mathcal{E}_1 \otimes_{\mathcal{O}_X} \Omega_X^1 \\ \downarrow \varphi & & \downarrow \varphi \otimes 1 \\ \mathcal{E}_2 & \xrightarrow{\nabla_2} & \mathcal{E}_2 \otimes_{\mathcal{O}_X} \Omega_X^1 \end{array}$$

commutes, as morphisms between $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$.

Here are some algebraic construction one can obtain with connections :

Proposition I.7.

Let $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$ two connections on X . The followings are also connections :

- 1) $(\mathcal{E}_1^*, (- \otimes 1) \circ \nabla_1)$,
- 2) $(\mathcal{E}_1 \oplus \mathcal{E}_2, \nabla_1 \oplus \nabla_2)$,
- 3) $(\mathcal{E}_1 \otimes_{\mathcal{O}_X} \mathcal{E}_2, \nabla_1 \otimes 1 + 1 \otimes \nabla_2)$
- 4) $(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}_1, \mathcal{E}_2), \phi \mapsto (\nabla_2 \circ \phi) - (\phi \otimes 1) \circ \nabla_1)$

II The Riemann-Hilbert correspondence

II.1 A correspondence in algebraic topology

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II.2 Holomorphic version on a complex manifold

Some ideas of this paragraph are taken from [[Voi02], §9.2], [[Sza12], §2.7] and [[Poo25], §7.7].

Theorem II.1 (Riemann-Hilbert correspondence, complex manifold case).

The fonctor

$$\begin{aligned} F : \text{LocFree}^\nabla(X) &\longrightarrow \text{LocSys}(X) \\ (\mathcal{E}, \nabla) &\longmapsto \mathcal{E}^\nabla := \text{Ker}(\nabla) \end{aligned}$$

induces an equivalence of categories.

It's inverse is given by

$$\begin{aligned} G : \text{LocSys}(X) &\longrightarrow \text{LocFree}^\nabla(X) \\ \mathcal{A} &\longmapsto (\mathcal{O}_X \otimes_{\mathbb{C}} \mathcal{A}, d \otimes 1) \end{aligned}$$

PROOF

Step 1 : G is well defined, that is : if $\mathcal{A} \in \text{LocSys}(X)$, then $(\mathcal{E}, \nabla) := (\mathcal{A} \otimes_{\mathbb{C}} \mathcal{O}_X, 1 \otimes d) \in \text{LocFree}^\nabla(X)$.

First remark that $d \otimes 1$ is well defined by extension of $\mathcal{O}_X \rightarrow \Omega_X^1$ with the tensor product. We also use the identification

$$(\mathcal{A} \otimes_{\mathbb{C}} \mathcal{O}_X) \otimes_{\mathcal{O}_X} \Omega_X^1 \simeq \mathcal{A} \otimes_{\mathbb{C}} (\mathcal{O}_X \otimes_{\mathcal{O}_X} \Omega_X^1) \simeq \mathcal{A} \otimes_{\mathbb{C}} \Omega_X^1$$

by super-commutativity of the tensor product.

For a small open set such that $\mathcal{A}|_U \simeq \mathbb{C}^r$, we have $(\mathcal{A} \otimes \mathcal{O}_X)|_U \simeq \mathcal{A}|_U \otimes \mathcal{O}_U \simeq \mathbb{C}^r \otimes \mathcal{O}_U \simeq \mathcal{O}_U^{\oplus r}$. Finally, we have :

$$\nabla^{(1)} \circ \nabla(a \otimes f) = \nabla^{(1)}((a \otimes 1) \otimes df) = \nabla(a \otimes 1) \wedge df + a \otimes d(df) = (a \otimes d1) \wedge df = 0.$$

We again used that the image of $a \otimes f$ in $\mathcal{E} \otimes_{\mathcal{O}_X} \Omega_X^1$ is identified with $(a \otimes 1) \otimes df$.

Step 2 : G is a functor.

If $\phi : A \rightarrow B$ is a morphism in $\text{LocSys}(X)$, we have an induced $G(\phi) = \phi \otimes 1 : A \otimes_{\mathbb{C}} \mathcal{O}_X \rightarrow B \otimes_{\mathbb{C}} \mathcal{O}_X$ that makes the following commutes :

$$\begin{array}{ccc} A \otimes_{\mathbb{C}} \mathcal{O}_X & \xrightarrow{\phi \otimes 1} & B \otimes_{\mathbb{C}} \mathcal{O}_X \\ 1 \otimes d \downarrow & & \downarrow 1 \otimes d \\ A \otimes_{\mathbb{C}} \Omega_X^1 & \xrightarrow{\phi \otimes 1} & B \otimes_{\mathbb{C}} \Omega_X^1 \end{array}$$

Step 3 : F is well defined, that is : if $(\mathcal{E}, \nabla) \in \text{LocFree}^\nabla(X)$, then $\mathcal{E}^\nabla \in \text{LocSys}(X)$.

This is the most difficult part. Let's sketch out the proof.

The result is equivalent to saying that *locally*, the solutions of our system of differential equations given by ∇ is a complex vector space, with dimension equal to the rank of the system. One can then think of the Cauchy-Lipschitz (or Picard-Lindelöf) theorem. It indeed works when X is a curve because we have locally an ordinary differential equation. If $n > 1$, this theorem cannot be applied.

There is an alternative for this type of system : the Frobenius theorem. One corollary of this theorem is the following lemma, that implies immediately the result.

Lemma II.2.

Let $x \in X$.

There exists an open set U containing x such that the evaluation $\text{ev}_x : \text{Ker}(\nabla|_U) \rightarrow \pi^{-1}(x)$ is an isomorphism (of sheaves of complex vector spaces).

Step 4 : F is a functor.

If $\phi : (\mathcal{E}_1, \nabla_1) \rightarrow (\mathcal{E}_2, \nabla_2)$ a morphism in $\text{LocFree}^\nabla(X)$, then it induces a morphism of sheaves $\phi : \text{Ker}(\nabla_1) \rightarrow \mathcal{E}_2$, and its image is a subset of $\text{Ker}(\nabla_2)$ by the relation $\nabla_2 \circ \phi = (\phi \otimes 1) \circ \nabla_1$.

Step 5 : $GF \simeq 1_{\text{LocFree}^\nabla(X)}$.

Let $(\mathcal{E}, \nabla) \in \text{LocFree}^\nabla(X)$. Set

$$\begin{aligned} \alpha_{(\mathcal{E}, \nabla)} : \text{Ker}(\nabla) \otimes_{\mathbb{C}} \mathcal{O}_X &\longrightarrow \mathcal{E} \\ s \otimes f &\longmapsto fs \end{aligned}$$

It is clearly well defined and for $s \otimes f \in \text{Ker}(\nabla) \otimes_{\mathbb{C}} \mathcal{O}_X$, we have, using the Liebniz rule :

$$\nabla(\alpha_{(\mathcal{E}, \nabla)}(s \otimes f)) = \nabla(fs) = s \otimes df = \alpha(s \otimes 1) \otimes df = (\alpha \otimes 1)((1 \otimes d)(s \otimes f)),$$

so $\alpha_{(\mathcal{E}, \nabla)}$ define a morphism in the category $\text{LocFree}^\nabla(X)$.

Better, it defines a natural transformation. Let $\phi : (\mathcal{E}_1, \nabla_1) \rightarrow (\mathcal{E}_2, \nabla_2)$ a morphism in $\text{LocFree}^\nabla(X)$. Then,

$$\phi(\alpha_{(\mathcal{E}_1, \nabla_1)}(f \otimes s)) = \phi(fs)$$

and

$$\alpha_{(\mathcal{E}_2, \nabla_2)}(GF(\phi)) = \alpha_{(\mathcal{E}_2, \nabla_2)}(F(\phi)(s) \otimes f) = f\phi(s).$$

This is equal because ϕ is a morphism of $\mathcal{O}_X$ -modules.

To show that this is an isomorphism, let U be a small open neighborhood of x , such that $\text{Ker}(\nabla|_U) \simeq \mathbb{C}^r$. Let $(s_1, \dots, s_r)$ a basis of $\text{Ker}(\nabla|_U)$. Then, $(s_1 \otimes 1, \dots, s_r \otimes 1)$ is a basis of $\text{Ker}(\nabla|_U) \otimes_{\mathbb{C}} \mathcal{O}_U$.

Moreover by taking U small enough, $(s_1, \dots, s_r)$ is also a $\mathcal{O}_U$ -basis of $\mathcal{E}(U) \simeq \mathcal{O}_U^{\oplus r}$. Indeed, the determinant of the matrix (in any basis of $\mathcal{E}_U$) with j -th column $(s_j^{(i)})^T$ is not zero at x , so does not vanish in a small neighborhood of x .

Therefore, the matrix of $\alpha_{(\mathcal{E}, \nabla)}$ in the basis $(s_1, \dots, s_r)$ (on both sides) is the identity, and is invertible.

Thus, $\alpha_{(\mathcal{E}, \nabla)}$ is invertible on each stalk, and it is an isomorphism of sheaves.

Step 6 : $FG \simeq 1_{\text{LocSys}(X)}$.

We clearly have for a local system $\mathcal{A}$ on X , $(\mathcal{A} \otimes_{\mathbb{C}} \mathcal{O}_X)^{1 \otimes d} \simeq \mathcal{A} \otimes_{\mathbb{C}} \mathbb{C} \simeq \mathcal{A}$, and it behaves well with morphisms, such that it defines an isomorphism of functors. $\square$

II.3 Meromorphic version on a complex manifold

For a Riemann surface (meaning, $\dim(X) = 1$), the previous result gives us that any finite dimensional representation of $\pi_1(X, x_*)$ can be obtained as the monodromy representation of some linear system of holomorphic differential equations.

21st Hilbert problem : given of finite set of points

II.3.1 In dimension 1

a. Elements of Fuchs theory

We focus our study around each point of S . Without loss of generality, the case around $z = 0$ is sufficient. We will denote D_ε the disk centered at 0 of radius ε and D_ε^* the associated punctured disk at $z = 0$.

If we want to translate the definition of a connection on D_ε , we would say that such a connection is a locally free sheaf of modules $\mathcal{E}$ over the sheaf of rings of meromorphic function defined on D_ε with only a pole at 0, equipped with a map $\nabla : \mathcal{E} \rightarrow \mathcal{E} \otimes_{\mathcal{O}_{D_\varepsilon}} \Omega_{D_\varepsilon}^1$ with linearity and Liebniz properties. We can assume that $\mathcal{E}$ is free. Moreover, it's more convenient to work with the field of germs of meromorphic functions around 0 denoted K .

Having localised our study, we then won't work with sheaf in this paragraph. We basically follow Haefliger's lecture in [Bor+87], and the theory exposed is usually referred to as the *Fuchs theory*. This shift of perspective leads us to the following definition.

Definition II.1.

A (germ of) *meromorphic connexion* at 0 is a finite dimensional vector space M over K equipped with an operator $\mathbb{C}$ -linear $\nabla : M \rightarrow M$ satisfying the Liebniz rule :

$$\forall (h, u) \in K \times M : \nabla(hu) = \frac{dh}{dz}u + h\nabla u.$$

With this definition, a morphism of connection is of linear morphism commuting with connections.

Remark II.1.

The ring $D[z^{-1}]$ of germs of differential operators with meromorphic coefficients acts on a meromorphic connection (M, ∇) by $Pu = \sum_{j=1}^N a_j \nabla^j u$ if $P = \sum_{j=1}^N a_j \frac{d^j}{dz^j}$ and $u \in M$.

Remark II.2.

As explained in I.16, given a basis of M , a connection map on M is equivalent to a system of differential equations : a meromorphic connection is given by $\nabla = d - Adz$, where A is a matrix with coefficient in K , and vice versa.

A morphism between systems of equation given by matrices A and B respectively is then

a matrix $M : D_\varepsilon^* \rightarrow M_n(K)$ such that $\frac{dM}{dz} = BM - MA$, and we say that two systems of differential equations are *equivalent* if $M(z)$ is invertible for any $z \in D_\varepsilon$ (or equivalently, the associate connections are isomorphic).

In the rest of the paragraph, we will juggle between these two definitions.

Let us compute the monodromy associated to a differential system $y' = Ay$ on D_ε .

First, recall that the universal cover of D_ε^* given by the exponential is $\{t \in \mathbb{C}, |e^{2i\pi t}| < \varepsilon\}$. Multi-valued functions are thus defined on this universal cover. t and $\tilde{\cdot}$ will belong to the universal cover and z to D_ε .

Let $\tilde{S}(t) = (\tilde{U}_1, \dots, \tilde{U}_n)$ a fundamental system of multi-valued solutions. A is 1-periodic on the universal cover, so $t \mapsto \tilde{S}(t+1)$ is also a solution. This, there exists a constant invertible matrix C such that $\tilde{S}(t+1) = \tilde{S}(t)C$.

Definition II.2.

The matrix C defined above is called the *monodromy matrix* of the system.

To get back in the punctured disk, let Γ a matrix such that $C = \exp(2i\pi\Gamma)$. Then, $\tilde{S}(t) \exp(-2i\pi t\Gamma)$ is 1-periodical on the universal cover, and one can well define $\Sigma(z) = \tilde{S}(\frac{1}{2i\pi} \log(z)) e^{\log(z)\Gamma}$. Finally, we set $S(z) = \Sigma(z) e^{\log(z)\Gamma}$.

One cannot expect to have a correspondence between the category of meromorphic connections and the one of representations of $\pi_1(D_\varepsilon^*)$, as the following example shows :

Example II.3.

Define the two connections $\nabla_1 = d$ and $\nabla_2 = d - \frac{dz}{z^2}$. They are not equivalent since $z \mapsto 0$ is not invertible, but they both have the trivial monodromy (the solutions of ∇_2 are of the form $Ce^{-1/z}$).

This phenomenon can be intuited as resulting of a *bad* behavior of the solutions at $z = 0$, meaning that they have there sort of an essential singularity.

Thus, we introduce a *nice* class of functions.

Definition II.3.

Let f be a multivalued function around 0.

f has *moderated growth* if, for all sector $S_{(a,b)}^\eta := \{z \in D_\eta^* \mid \arg(z) \in (a,b)\}$ over which f is defined, there exist $k \in \mathbb{N}$ and $C > 0$ such that :

$$\forall z \in S_{(a,b)}^\eta, |f(z)| \leq C|z^{-k}|.$$

Example II.4.

- 1) If f is a single-valued function, f has moderate growth if, and only if f is meromorphic on D_ε (for ε small enough).
- 2) z^α for $\alpha \in \mathbb{C}$, $\log(z)^m$ for $m \in \mathbb{N}$, have moderate growth.

The following result is very powerful : it gives you information on the behavior of the solutions of a differential equation just by checking a condition on the coefficients of the differential operator with a very simple computation.

Theorem II.3 (Fuchs).

Let $P = \sum_{k=0}^n a_k(z) \frac{d^{n-k}}{dz^{n-k}}$ a differential operator on D_ε with the a_k 's meromorphic. The followings are equivalent :

- i) the solutions of $Pu = 0$ have moderate growth,
- ii) for every $1 \leq i \leq n$, $\frac{a_i}{a_0}$ has a pole of order at most i at 0.

These systems are those we are interested in for an expected correspondence.

Definition II.4.

A differential operator is said to be *regular* if it satisfies the conditions of the previous theorem.

Example II.5.

As expected, the example given in II.3 is not regular.

Theorem II.4.

Let $(E) : y' = Ay$ a differential system on D_ε . The followings are equivalent :

- i) (E) is equivalent to a regular system,
- ii) (E) is equivalent to a system $y' = \frac{B(z)}{z}y$ for some matrix B holomorphic on D_ε ,
- iii) (E) is equivalent to a system $y' = \frac{\Gamma}{z}y$ for some constant matrix Γ .

PROOF

.

□

With this theorem, the equivalent definition in terms of connections is the following :

Definition II.5.

Let (M, ∇) be a meromorphic connection.

It is said to be a *regular connection* if there exists a basis $(e_1, \dots, e_n)$ of M over K and holomorphic functions $(b_{i,j}(z))$ such that :

$$\forall i \in \llbracket 1, n \rrbracket, \forall z \in D_\varepsilon, z \nabla e_i = - \sum_{j=1}^n b_{i,j}(z) e_j.$$

The following theorem is what we were aiming for.

Theorem II.5.

Two regular systems on D_ε are equivalent if, and only if, their monodromy is conjugated.

PROOF

.

□

Stated like that, it is not as precise as we would have wanted for a equivalence of categories like in II.2. The next paragraph will make it more accurate, based on the theorem above.

First, a result on the good behavior of algebraic construction (see I.7) with regularity :

Proposition II.6 ([HTT08], Proposition 5.1.11).

Let M and N be regular meromorphic connections.
Then $\text{Hom}_K(M, N)$ and $M \otimes_K N$ are regular.

PROOF

.

□

b. The local lifting

This paragraph is partially inspired by [[Sza12], §2.7] and [[HTT08], Lemma 5.2.19] for the lifting of morphisms.

The precise Riemann-Hilbert correspondence will be stated next paragraph, based on what we already did in the holomorphic case. In this one, we try to construct a unique (regular) meromorphic connection on D_ε given a holomorphic connection on D_ε^* .

We return to the language of sheaves. Indeed, according to a general fact on non-compact Riemann surfaces, a locally free $\mathcal{O}_{D_\varepsilon^*}$ -module on D_ε is actually free (for example, see [[For81], Theorem 30.4]). We will denote by $\mathcal{O}_{D_\varepsilon}(*0)$ the sheaf of meromorphic functions on D_ε with a pole at $z = 0$.

Proposition II.7.

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on D_ε^* .
There exists regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on D_ε which agrees with $(\mathcal{E}, \nabla)$ on D_ε .
This connection is unique up to isomorphisms (of meromorphic connections).

PROOF

We can assume $\mathcal{E} = \mathcal{O}_{D_\varepsilon^*}^r$ and $\nabla = d - Adz$. Set $S(z)$ a fundamental system of solutions. Let C the monodromy matrix associated to ∇ , $S(z)$ a fundamental system of solutions, and let B a matrix such that $C = \exp(2i\pi B)$. We consider the connection $\bar{\nabla} = d - \frac{B}{z}dz$. A fundamental system of solutions is given by $z^B = e^{B \log(z)}$.

Let $U(z) = S(z)z^{-B}$. Then

$$U(ze^{2i\pi}) = S(ze^{2i\pi})e^{-B \log(ze^{2i\pi})} = S(z)Ce^{-2i\pi B}z^{-B} = U(z).$$

Here, by an abuse of notation, $ze^{2i\pi}$ denotes the composition by the simple loop going around 0 counter-clockwise.

Therefore, $U(z)$ is single-valued, and then holomorphic on D_ε^* . Its inverse is too, so it defines a isomorphism between $(\mathcal{E}, \nabla)$ and $(\mathcal{O}_{D_\varepsilon}(*0)^r, \bar{\nabla})$.

Uniqueness comes from II.5.

□

To make a functorial statement, we need additionally an extension of morphisms, we will directly do it for the global framework.

c. Global study via patching

We will state the "meromorphic" Riemann-Hilbert on $\mathbb{P}_{\mathbb{C}}^1$ minus a finite set of points. We firstly introduce some global notation compared with what we did the last two paragraphs, and then state and proof the result.

Some ideas comes from [Sza12].

In this paragraph, we set $X = \mathbb{P}_{\mathbb{C}}^1$, $D = \{s_1, \dots, s_k\} \subset X$ and $Y = X \setminus D$.

We denote by $\mathcal{O}_X(*D)$ the sheaf of meromorphic functions on X with poles along D .

Definition II.6.

Let $\bar{\mathcal{E}}$ a locally free $\mathcal{O}_X(*D)$ -modules.

A *meromorphic connection* on $\bar{\mathcal{E}}$ is a $\mathbb{C}$ -linear map $\bar{\nabla} : \bar{\mathcal{E}} \rightarrow \bar{\mathcal{E}} \otimes_{\mathcal{O}_X(*D)} \Omega_X^1$ satisfying the Liebniz rule.

Definition II.7.

We define the category $\text{LocFree}^{\bar{\nabla}}(X, D)$ with

- locally free $\mathcal{O}_X(*D)$ -modules equipped with a meromorphic (flat) connection as objects,
- morphisms of $\mathcal{O}_X(*D)$ -modules commuting with the connections as morphisms.

Theorem II.8.

Let $(\mathcal{E}, \nabla)$ a holomorphic connection on Y .

There exists regular meromorphic connection $(\bar{\mathcal{E}}, \bar{\nabla})$ on X which agrees with $(\mathcal{E}, \nabla)$ on D_{ε} .

This connection is unique up to isomorphisms (of meromorphic connections).

PROOF

Let D_i small disks around each s_i that do not meet. On each D_i , let $(\bar{\mathcal{E}}_i, \bar{\nabla}_i)$ the unique (up to isomorphisms) lifting of $(\mathcal{E}_i, \nabla_i) := (\mathcal{E}, \nabla)|_{D_i}$ given by II.7. On an open set U containing the complementary of $\bigcup_{i=1}^k D_i$ that does not meet S , we set $\bar{\mathcal{E}} = \mathcal{E}$.

Then, we can glue back the sheaves on $X = \bigcup_{i=1}^k D_i \cup U$ by I.4 (the isomorphisms $\theta_{i,j}$ are just given by the ones coming from the construction of the $\bar{\mathcal{E}}_i$, and the triple intersection condition is empty), which gives us the sheaf $\bar{\mathcal{E}}$ on X that agrees with $\mathcal{E}$ on Y .

□

Now, we deal with morphisms :

Proposition II.9 ([HTT08], Lemma 5.2.19).

Let $\phi : (\mathcal{E}_1, \nabla_1) \rightarrow (\mathcal{E}_2, \nabla_2)$ a morphism between holomorphic connections on Y .

Then, there exists a morphism of meromorphic connections $\bar{\phi} : (\bar{\mathcal{E}}_1, \bar{\nabla}_1) \rightarrow (\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ on X that restricts to ϕ .

We need two intermediate result.

Lemma II.10.

The restriction map given by II.8 induces a bijection :

$$\Gamma(X, \text{Ker}(\bar{\nabla})) \xrightarrow{\sim} \Gamma(Y, \text{Ker}(\nabla)).$$

PROOF

The injectivity is clear. For the surjectivity, let $s \in \Gamma(Y, \text{Ker}(\nabla))$. $\bar{\nabla}$ is regular, so s , a solution of the differential equation induced is single-valued as being defined on the whole of Y , and therefore has moderate growth - meaning that it can be extended to a meromorphic section on D . This concludes. $\square$

This one follows the definition.

Lemma II.11.

Let $(\bar{\mathcal{E}}_1, \bar{\nabla}_1)$ and $(\bar{\mathcal{E}}_2, \bar{\nabla}_2)$ two connections on D_ε .
Then,

$$\Gamma(X, \mathcal{H}om_{\mathcal{O}_X(*D)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)^\nabla) \simeq \text{Hom}_{\text{LocFree}^{\bar{\nabla}}(X,D)}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)),$$

where the ∇ on the left-hand side is the one associated to the sheaf $\mathcal{H}om$ (see I.7).

PROOF (??)

Set $\mathcal{E} = \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{E}_1, \mathcal{E}_2)$ and $\bar{\mathcal{E}} = \mathcal{H}om_{\mathcal{O}_X(*S)}(\bar{\mathcal{E}}_1, \bar{\mathcal{E}}_2)$ with the associate ∇ and $\bar{\nabla}$. By II.6, $\bar{\mathcal{E}}$ is regular, so we can apply the first lemma. The second one thus gives a bijection :

$$\text{Hom}((\bar{\mathcal{E}}_1, \bar{\nabla}_1), (\bar{\mathcal{E}}_2, \bar{\nabla}_2)) \simeq \text{Hom}((\mathcal{E}_1, \nabla_1), (\mathcal{E}_2, \nabla_2)),$$

the surjectivity of which ensures the result. $\square$

Theorem II.12.

The restriction functor

$$\text{LocFree}^{\bar{\nabla}}(X, D) \rightarrow \text{LocFree}^{\nabla}(Y)$$

induces an equivalence of categories.

PROOF

The pseudo-inverse functor is given by the construction above, and all the other verifications are clear. $\square$

With the work done the previous subsection, we have the correspondence as wished :

Corollary II.13 (Riemann-Hilbert correspondence, meromorphic case, dimension 1).

There is an equivalence of categories :

$$\mathrm{LocFree}^{\bar{\nabla}}(X, D) \cong \mathrm{LocSys}(Y).$$

II.3.2 Sketch of a proof in higher dimension

bla

II.4 Meromorphic version on a smooth variety

III Periods

for introduction : picard fuchs equation : from periods depending of a parameter, construct a differential equations attached

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